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Started	Sunday, 4 October 2026, 8:42 PM
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Duration	8 mins 4 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

a. n/2
b. log n
c. n ✓
d. 1

Question **2**

Correct

Mark 1.00 out of 1.00

What type of search would be most appropriate for finding an element in a list that is frequently updated?

a. Hash search
b. Interpolation search
c. Linear search ✓
d. Binary search

Question **3**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

a. O(1)
b. O(log n)
c. O👎 ✓
d. O(n log n)

Question **4**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

- a. Checking each element sequentially ✓
- b. Sorting the list before searching
- c. Skipping every second element
- d. Dividing the list in half repeatedly

Question **5**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

- a. When the list is small and sorted
- b. When the list is small and unsorted ✓
- c. When the list is large and sorted
- d. When the list is large and unsorted

Question **6**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

- a. It requires the list to be sorted ✓
- b. It is slower than linear search for small lists
- c. It can only be applied to large lists
- d. It does not work with negative numbers

Question **7**

Correct

Mark 1.00 out of 1.00

_____ search takes a sorted/ordered list and divides it in the middle.

- a. Linear
- b. Binary ✓
- c. Hash
- d. Both (1) & (3)

Question **8**

Correct

Mark 1.00 out of 1.00

_____sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements in case they are unordered in n-1 passes.

a. Bubble ✓

b. Complexity

c. Insertion

d. Selection

Question **9**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

a. It always starts from the beginning of the list

b. It compares elements sequentially

c. It can be applied only if the list is sorted ✓

d. It works on unsorted lists

Question **10**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

a. Binary search can find multiple instances of the target element

b. Binary search does not require dividing the list

c. Binary search has a lower time complexity on large, sorted lists ✓

d. Binary search works on unsorted lists

Question **11**

Correct

Mark 1.00 out of 1.00

Which of the following is a conventional searching technique?

- a. Dynamic search
- b. Binary search
- c. Hashing
- d. Linear search ✓

Question **12**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

- a. Linear search ✓
- b. Binary search
- c. Hash search
- d. Both (1) & (3)

Question **13**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

- a. 100
- b. 1000
- c. 500
- d. 10 ✓

Question **14**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

- a. When the list is sorted ✓
- b. When the list is very small
- c. When the list is unsorted
- d. When the list contains duplicate elements

Question **15**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

- a. There must be a mechanism to access middle element directly
- b. Binary search algorithm is not efficient when the data elements more than 1500 ✓
- c. Must use a sorted array
- d. Requirement of sorted array is expensive when a lot of insertion and deletions are needed